

Q.22 State two advantages of automation in haematology.

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SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Explain the formation, functions and degradation of haemoglobin in detail.

Q.24 Describe the methods of counting RBC, WBC and platelets using a haemocytometer. Add reference values and common errors.

Q.25 Write a detailed note on differential leucocyte counting (DLC): preparation and staining of the blood film, normal values and clinical significance.

1st Year / MLT Subject :Haematology

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 The globin part of haemoglobin is composed of:

- a) Lipids b) Proteins
- c) Carbohydrates d) Nucleic acids

Q.2 Which type of haemoglobin is abnormal and seen in sickle cell disease?

- a) HbA b) HbF
- c) HbS d) HbC

Q.3 Which method is considered the gold standard for haemoglobin estimation?

- a) Sahli's method
- b) Cyanmethaemoglobin method
- c) Tallquist method
- d) Gasometric method

- Q.4 The chamber used for manual counting of blood cells is:
- Neubauer chamber
 - C-Chamber
 - KOH chamber
 - Howard chamber
- Q.5 The most abundant WBC in a normal blood smear is :
- Lymphocyte
 - Neutrophil
 - Eosinophil
 - Monocyte
- Q.6 Automated blood cell counters commonly work on the principle of:
- ELISA
 - Coulter's principle (Electrical impedance)
 - Nephelometry
 - Immunodiffusion

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 The prosthetic group present in haemoglobin is _____.
- Q.8 HbF is the major haemoglobin present in the _____.
- Q.9 The dilution fluid used in RBC counting is _____.

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- Q.10 The stain commonly used for DLC is _____.
- Q.11 Accuracy refers to the closeness of results to the _____ value.
- Q.12 In an automated cell counter, the number of cells is counted as they pass through an aperture with an applied ____.

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 Define Blood and state its two main functions.
- Q.14 Write two differences between HbA and HbF.
- Q.15 Mention two advantages of cyanmethaemoglobin method over Sahli's method.
- Q.16 Name any two counting chambers and their uses.
- Q.17 Write the calculation formula for WBC count using a Neubauer chamber.
- Q.18 List two common errors in haemocytometry and how to prevent them.
- Q.19 Give the normal DLC values of neutrophils and lymphocytes.
- Q.20 Write the morphological features of eosinophils and basophils in a peripheral smear.
- Q.21 Differentiate between precision and accuracy in quality control.

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